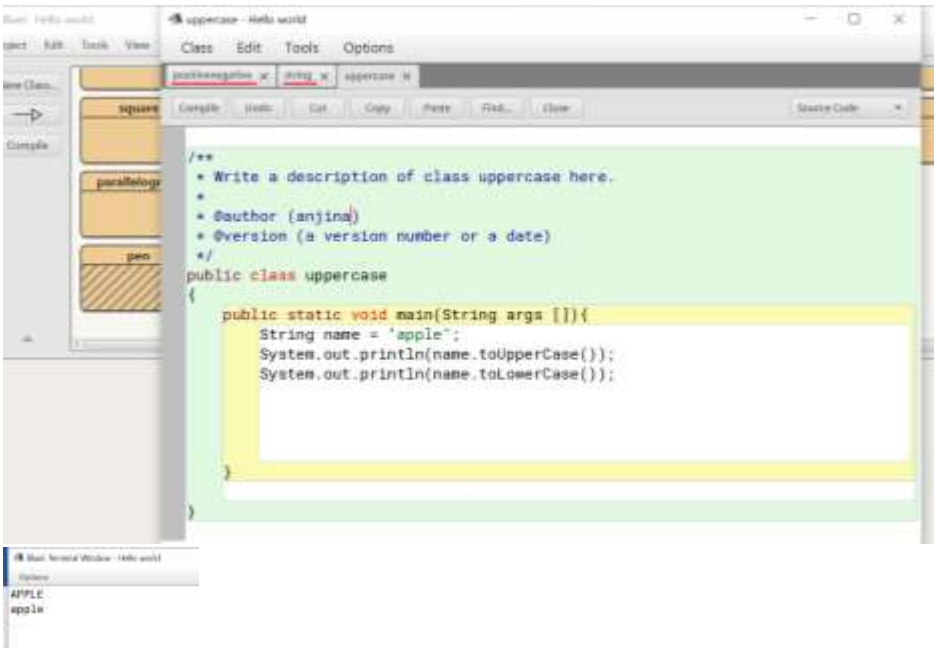
Group A

1. What is the String class in Java? Is String a data type?
   Strings are sequence of characters. Yes, string is a data type.

2. How can you make a String upper case or lower case in Java?
   The toUpperCase() is used to convert string to upper case and toLowerCase() is used to convert to lower case.



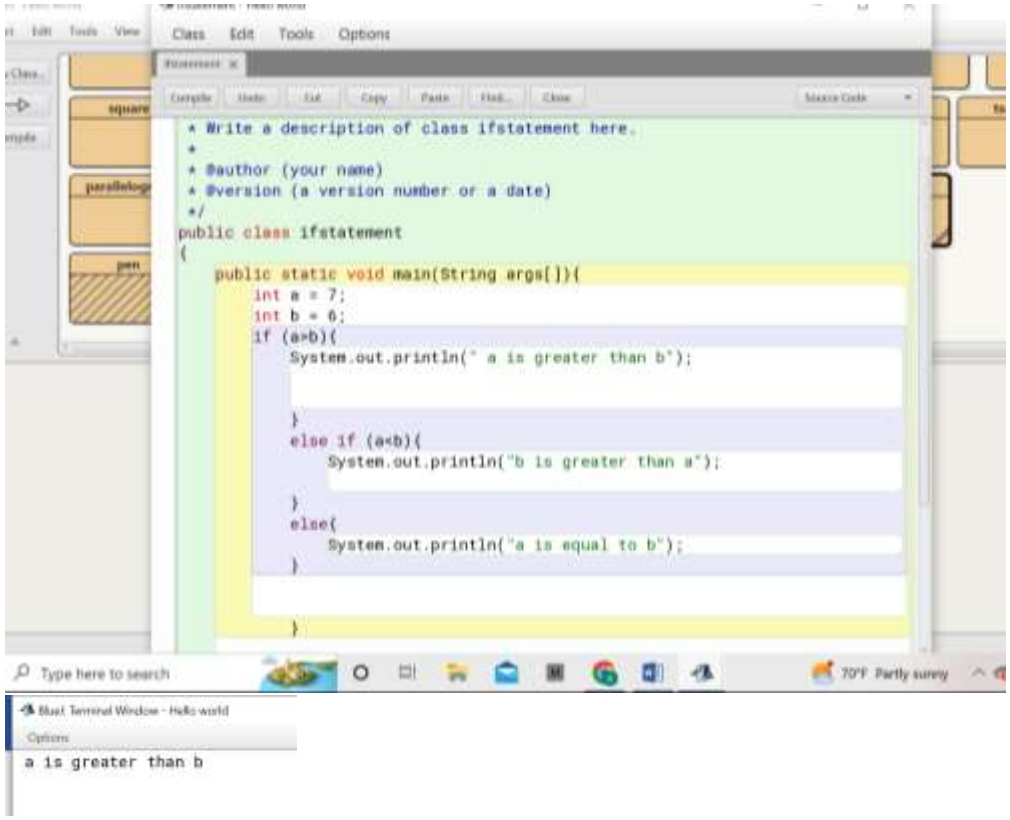
3. Can you use String in switch case in Java? Explain it briefly.
   Yes, we can use string in switch case in Java. It tests the equality of variable against multiple values.

4.Explain different types of conditional statement in java.

The different types of conditional statements in Java are:
If: If is used to specify a block of code to be executed, if the specified condition is true.

If-else: if-else is used to specify a block of code to be executed, if the first condition is false.

Else: Else is used to specify a block of code to be executed, if the same condition is false.



5) What is the value of the variable num after the following is executed?

o int k = 5;

o int num = 0;

o int num1 = num + k * 2;

o int num2 = num + k * 2;

variablenum - Hello world

Class Edit Tools Options

```java
/**
 * Write a description of class variablenum here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class variablenum
{
    public static void main (String []args){
        int k = 5;
        int num = 0;
        int num1 = num+k*2;
        int num2 = num+k*2;
        System.out.println(num1);
    }
}
```

Blue Terminal Window - Hello world

```
10
```

6. How do you split a string in Java?

split - Hello world

Class Edit Tools Options

```java
/**
 * Write a description of class split here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class split
{
    public static void main(String [] args){
        String str = "as,df,_jkl";
        String[] arr = str.split(",");
        for (int i=0; i<arr.length; i++){
            System.out.println(arr[i]);
        }
    }
}
```

Blue Terminal Window - Hello world

```
as
df
_jkl
```

7. How do you check if two Strings are equal in Java?

To check if the strings are equal in Java we use equal method.

Group – B

1. Find the difference between Beth's age (57) and Tom's age (34).

```
/**
 * Write a description of class age here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class age
{
    public static void main (String[] args){
        int Beth = 57;
        int Tom = 34;
        int difference = Beth-Tom;
        System.out.println(difference);

    }
}
```

```
23
```

2. Develop a system to store your name as variable.

```
/**
 * Write a description of class variable here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class variable
{
    public static void main (String args []){
        String name = "Anjina";
        var a = name;
        System.out.println(a);

    }
}
```

```
Anjina
```

3. Create the above java program in the java environment and then modify the program to use the following statements. Note down the response to each. Do they differ from what you would expect?

```
/**
 * Write a description of class truee here.
 *
 * @author (anjina)
 * @version (a version number or a date)
 */
public class truee
{
    public static void main (String [] args ){
        int x = 20;
        System.out.println(x>10&&x<40);
    }
}
```

```
true
```

```java
/**
 * Write a description of class falsee here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class falsee
{
    public static void main (String [] args ){
        int x = 20;
        System.out.println(x>10&&x<60||x>40);
    }
}
```

Blue: Terminal Window - Hello world

Options

```
true
```

truefalse - Hello world

Class Edit Tools Options

Trues | falses | truefalse

Compile Undo Cut Copy Paste Find... Close

```java
/**
 * Write a description of class truefalse here.
 *
 * @author (anjina)
 * @version (a version number or a date)
 */
public class truefalse
{
    public static void main(String[] args){
        int x = 20;
        System.out.println(x>10&&x<60||x>40);
    }

}
```

Blue: Terminal Window - Hello world

Options

```
true
```

falsetrue - Hello world

Class Edit Tools Options

```java
/**
 * Write a description of class falsetrue here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class falsetrue
{
   public static void main (String [] args ){
       int x = 20;
       System.out.println(!(x>40)&&x<30);
   }
}
```

Blue: Terminal Window - Hello world

Options

```
true
```

booleanresult - Hello world

Class Edit Tools Options

```java
/**
 * Write a description of class booleanresult here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class booleanresult
{
    public static void main (String[] args ){
        int x= 20;
        System.out.println(!(x>40)&&!(x>50&&x>60));
    }
}
```

Blue: Terminal Window - Hello world

Options

```
true
```

4. Find the difference between 7 factorial and 5 factorial.

factorialdiff - Hello world

Class Edit Tools Options

loop | inputloop | factorialdiff

Compile Undo Cut Copy Paste Find... Close

```
 */
public class factorialdiff
{
    public static void main (String [] args){
        int fact1 = 1;
        int fact2 = 1;
        int i = 1;

        while(i<=7){
            fact1 = fact1 * i;
            i++;
        }
        System.out.println(fact1);
        i = 1;
        while(i<=5){
            fact2 = fact2 * i;
            i++;
        }
        System.out.println(fact2);
        int a = fact1-fact2;
        System.out.println(a);
```

Blue: Terminal Window - Hello world

Options

```
5040
120
4920
```

5. Complete the following questions by taking user input.

o Write a Java program that prompts a user for their last name and stores it in a variable named last_name.

lastname - Hello world

Class Edit Tools Options

loop | inputloop | factorialdiff | lastname | split | isoppositive | pelse

Compile Undo Cut Copy Paste Find... Close

```
/**
 * Write a description of class lastname here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class lastname
{
    public static void main (String [] args ){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter your last name: ");
        String last_name = sc.nextLine();
        System.out.println(last_name);

    }
}
```

```
Enter your last name:
gurung
gurung
```

Give an instruction that prompts a user for their age and stores it as an integer in a variable named age.

```
import java.util.*;
public class intage
{
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter your age: ");
        int age = sc.nextInt();
        System.out.println(age);


        }
    }
```

```
Enter your age:
27
27
```

Give an instruction that prompts a user for their temperature and stores it as a float in a variable named current_temperature.

```java
/**
 * Write a description of class temperature here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class temperature
{
    public static void main (String [] args){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter your temperature: ") ;
        float current_temperature = sc.nextFloat();
        System.out.println(current_temperature);

    }
}
```

```
Enter your temperature:
98.4
98.4
```

6. Give a call to printf that is provided one string that displays the following address on three separate lines:

```java
/**
 * Write a description of class johndoe here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class johndoe
{
    public static void main (String [] args){
        System.out.printf("John doe %n123dudley street %n123dudley street");

    }
}
```

```
John doe
123dudley street
123dudley street
```

7. Write a java program in which:

a) The user enters either 'A', 'B', or 'C'. If 'A' is entered, the program should display the word 'Apple'; if 'B' is entered, it displays 'Banana'; and if 'C' is entered, it displays 'Coconut'. Use nested if statements for this.

```
import java.util.*;
public class apple
{
    public static void main (String [] args){
        Scanner sc = new Scanner (System.in);
        String input;
        int a =1;
        int b = 1;
        int c= 1;
        while (a==1 || b==1 || c==1){
            System.out.println("Enter A or B or C");
            input = sc.nextLine();
            if (input.equals("A")){
                System.out.println("Apple");
                a=0;

            }else if (input.equals("B")){
                System.out.println("Banana");
                b=0;

            }else if (input.equals("C")){
                System.out.println("Coconut");
                c=0;
```

```
        a=0;

    }else if (input.equals("B")){
        System.out.println("Banana");
        b=0;

    }else if (input.equals("C")){
        System.out.println("Coconut");
        c=0;



    }
    else{
        System.out.println("Please try again");
    }
    }
}
}
```

```
BlueJ: Terminal Window - Hello world
Options
Enter A or B or C
c
Please try again
Enter A or B or C
C
Coconut
Enter A or B or C
A
Apple
Enter A or B or C
B
Banana
```

c) A student enters the number of college credits earned. If the number of credits is greater thanor equal to 90, 'Senior Status' is displayed; if greater than or equal to 60, 'Junior Status' is displayed; if greater than or equal to 30, 'Sophomore Status' is displayed; else, 'Freshman Status' is displayed.

```
Class   Edit   Tools   Options
johnsbw ×   apple ×   freshman ×
Compile   Undo   Cut   Copy   Paste   Find...   Close   Source Code

 * Write a description of class freshman here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class freshman
{
   public static void main (String [] args){
       Scanner sc = new Scanner (System.in);
       System.out.println("Enter your credit: ");
       int credit = sc.nextInt();
       if (credit>=90){
           System.out.println("Senior status");
       }else if (credit >=60){
           System.out.println("Junior status");
       }else if (credit>=30){
           System.out.println("Sophomore status");

       }else{
           System.out.println("Freshman");
       }

   }
}
```

```
BlueJ: Terminal Window - Hello world
Options
Enter your credit:
75
Junior status
```

Group-C

1. Create a Java software that will ask the user for a number and then display whether it is positive or negative.

```java
/**
 * Write a description of class positivenegative here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class positivenegative
{
    public static void main (String args[]){
        Scanner sc = new Scanner (System.in);
        System.out.print("enter a number: ");
        int number = sc.nextInt();
        if (number>0){
            System.out.println("positive");
                    }else{
                        System.out.println("negative");
                    }


        }

    }
```

```
enter a number: 8
positive
```

2. Your name left justified 15 spaces. [Formatted Output]

```java
/**
 * Write a description of class justified here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class justified
{
    public static void main (String [] args ){
        System.out.printf("%15s", "Anjina");
    }
}
```

3. Your name right justified 15 spaces. [Formatted Output]

4. There were bunch of students who were curious about their total marks, percentage and grade using the marks from five subjects as input. Develop a system to help them find their grades.

```java
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class result
{
    public static void main (String[] args ){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter your marks: ");
        double M= sc.nextDouble();
        if (M>=80){
            System.out.println("Your grade is A");

        }else if (M>60){
            System.out.println("Your grade is B");

        }else if (M>=40){
            System.out.println("Your garde is C");
        }else if (M>=20){
            System.out.println("Your grade is D");

        }else
        System.out.println("You failed");
    }
}
```

```
Enter your marks:
85
Your grade is A
```

5. Write a Java program that allows the user to enter two integer values and displays the results with the following arithmetic operators applied to them. For example, if the user enters the values 7 and 5, the output would be:

o Addition: 7 + 5 = 12

o Subtraction: 7 – 5 = 2

o Multiplication: 7 * 5 = 35

o Division: 7 / 5 = 1.40

o Modulus: 7 % 5 = 2

o Exponentiation: 7 ** 5 = 16,807

```
 * @version (a version number or a date)
 */
import java.util.*;
public class operators
{
    public static void main (String [] args){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter first number: ");
        int a = sc.nextInt();
        System.out.println("Enter second number: ");
        int b = sc.nextInt();
        int add = a+b;
        System.out.println(add);
        int sub = a-b;
        System.out.println(sub);
        int multiplication = a*b;
        System.out.println(multiplication);
        double division = a/b;
        System.out.println(division);
        double exponentiation = Math.pow(a,b);
        System.out.println(exponentiation);
        float modulus = a%b;
        System.out.println(modulus);
```

```
Enter first number:
5
Enter second number:
3
8
2
15
1.0
125.0
2.0
```

6. Let’s create a java program to input a number and check whether it is a Buzz number or not. A number is said to be a buzz number when it ends with 7 or is divisible by 7.

```java
/**
 * Write a description of class buzz here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class buzz
{
    public static void main (String [] args ){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter a number: ");
        int num = sc.nextInt();
        if (num%10==7){
            System.out.println("This is a buzz number" + num);
        }else if (num%7==0){
            System.out.println("The number is divisible by 7"+ num);
        }else{
            System.out.println("This is not a buzz number");
        }

    }

}
```

```
Enter a number:
7
This is a buzz number7
```

7. Let’s take an example program where we will take the age of user as input and find whether he is a child, adult, or senior on the basis of age. Using Java if-else-if ladder statements.

```java
/**
 * Write a description of class senior here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class senior
{
    public static void main (String [] args){
        Scanner sc = new Scanner (System.in);
        int n1 = sc.nextInt();
        if (n1<18){
            System.out.println("Child");

        }else if (n1<60){
            System.out.println("Adult");
        }else{
            System.out.println("Senior");
        }
    }
}
```

```
55
Adult
```

8. Bruno Mars just appeared his examination and got 75%. He goes to his tutor and asks his grade. Now being a tutor you need to develop a program which tells his grade.

```java
/**
 * Write a description of class Bruno here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class Bruno
{
    public static void main (String [] args ){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter your percentage: ");
        float percentage = sc.nextFloat();
        char grade = percentage>=90? 'A' : percentage >=80? 'B' : percentage >=70? 'C' : percentage >= 60? 'D': 'F';
        System.out.println("Grade: " + grade);
    }
}
```

```
Enter your percentage:
78
Grade: C
```

9. If a customer wants to take a t-shirt from your shop and he wants to buy a t-shirt and feeds in his/her size. Then print the availability as per their preference. [Using Switch Case Statement].

```
*/
import java.util.*;
public class tshirt
{
    public static void main (String [] args ){
        Scanner sc = new Scanner (System.in);
        System.out.println ("Enter your tshirt size : ");
        String size = sc.nextLine();
        String s,m,l;
        switch (size){
            case "s":
                System.out.println("Your size is available");
                break;
                case "m":
                    System.out.println("Your size is available");
                    break;
                    case "l":
                        System.out.println("Your size is available");
                        break;
                        default :
                            System.out.println("It is not available");

        }
    }
```

```
Enter your tshirt size :
m
Your size is available
```

Group D

1. Let’s create a printing application program where we will take the number of copies to be printed as input from the user and then prints the price per copy and the total price for the printing copies. The chart price to print the number of copies is given below:

- 0 – 99 : $0.30 per copy
- 100 – 499 : $0.28 per copy
- 500 – 799 : $0.27 per copy
- 800 – 1000 : $0.26 per copy
- over 1000 : $0.25 per copy

```java
{
    public static void main(String [] args ){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter the number of copies: ");
        int c = sc.nextInt();
        if (c>=0&&c<=99){
            System.out.println("Price per copy is $0.03");
            double p1 = 0.03*c;
            System.out.println("The total price is: "+p1);

        }else if (c>=100&&c<=599){
            System.out.println("Price per copy is $0.28");
            double p2 = 0.28*c ;
            System.out.println("The total price is : p2");

        }else if (c>=500&&c<=799){
            System.out.println("Price per copy is $0.27");
            double p3 = 0.27*c;
            System.out.println("The total price is: p3");

        }else if (c>=800&&c<=1000){
            System.out.println("Price of copy is $0.03");
            double p1 = 0.03*c;
            System.out.println("The total price is: p1");
```

```
Enter the number of copies:
20
Price per copy is $0.03
The total price is: 0.6
```

2. Follow the simulation of Floor example from lecture slide and develop a system where you need to ask user the floor number. Also determine whether the floor is actual floor or not.

```java
/**
 * Write a description of class floor here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
import java.util.*;
public class floor
{
   public static void main (String [] args ){
       Scanner sc = new Scanner (System.in);
       System.out.println("Enter the floor: ");
       int f = sc.nextInt();
       int af;
       if (f==13){
           af = f-1;

       }else {
           af=f;

       }
       System.out.println("The elevator will be in the actual floor" + af);
   }
}
```

```
Enter the floor:
6
The elevator will be in the actual floor6
```

3. [Scenario] You're waiting at a station and the announcer has just broadcast that your train is going to be 13445 seconds late. You need to work out in understandable terms what that means. You assume this is going to be quite a long time so you whip out your laptop to write a program to convert the seconds into hours, minutes and seconds, aiming to maximize readability by giving priority to the

largest units, i.e. the resulting seconds and minute's values must not be greater than 60. You will need four variables to hold: the total number of seconds; the number of hours; the number of minutes; and the number of remaining seconds. The example output should look something like this:

13442 Seconds is: 3 Hours, 44 Minutes and 5 Seconds.

hours - Hello world

Class Edit Tools Options

```java
/**
 * Write a description of class hours here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class hours
{
    public static void main (String [] args ){
        int n = 13445;
        int hours = n/3600;
        int remSecWithoutHours = n -(hours*3600);
        int minutes = remSecWithoutHours / 60;
        int seconds = remSecWithoutHours -(minutes*60);
        System.out.println(n+'seconds = " + hours + "hours" + minutes +'minutes" + seconds + 'second');

    }
}
```

Blue: Terminal Window - Hello world

Options

```
13445seconds = 3hours44minutes5second
```